

SynthRCT: Scalable Conditional Deformation Synthesis for Synthetic Repeat CT Generation

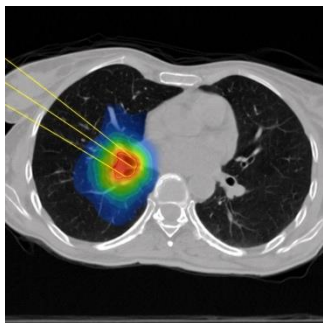
Tomas Guija-Valiente, Blanca Rodriguez-Gonzalez, Norberto Malpica

INTRODUCTION

Anatomical variations in proton therapy



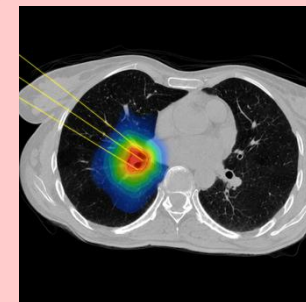
PLANNING CT



DURING TREATMENT



CONSEQUENCE

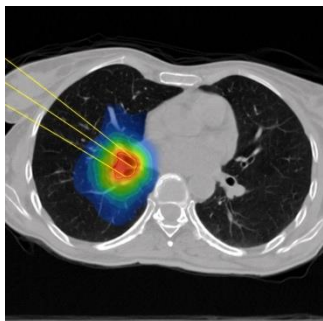


Dose may miss target or
overdose organs at risk

Anatomical variations in proton therapy



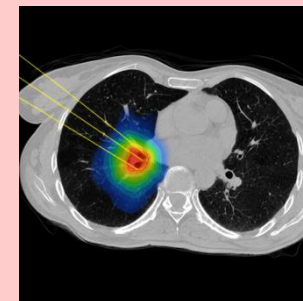
PLANNING CT



DURING TREATMENT



CONSEQUENCE

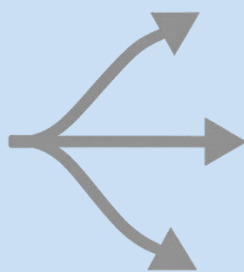


Dose may miss target or overdose organs at risk

CURRENT PRACTICE FOR ROBUSTNESS EVALUATION



Planning CT



Simple shifts



Scaling

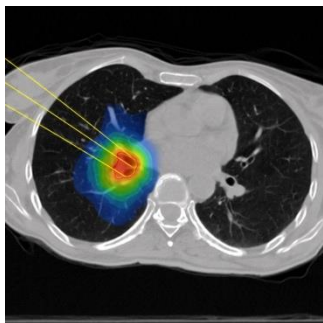


Predefined scenarios

Anatomical variations in proton therapy



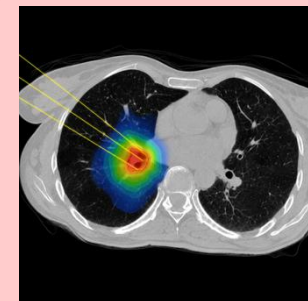
PLANNING CT



DURING TREATMENT

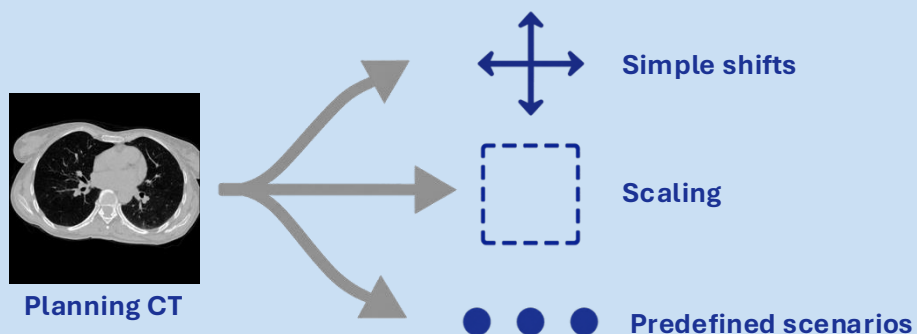


CONSEQUENCE

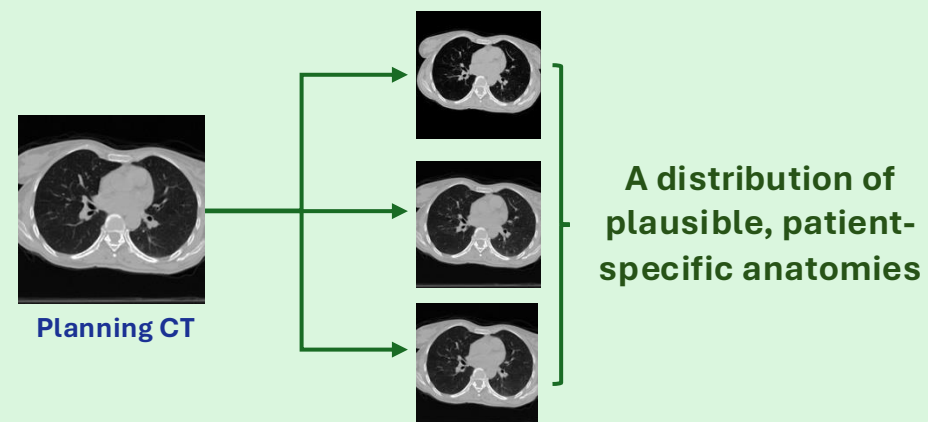


Dose may miss target or overdose organs at risk

CURRENT PRACTICE FOR ROBUSTNESS EVALUATION



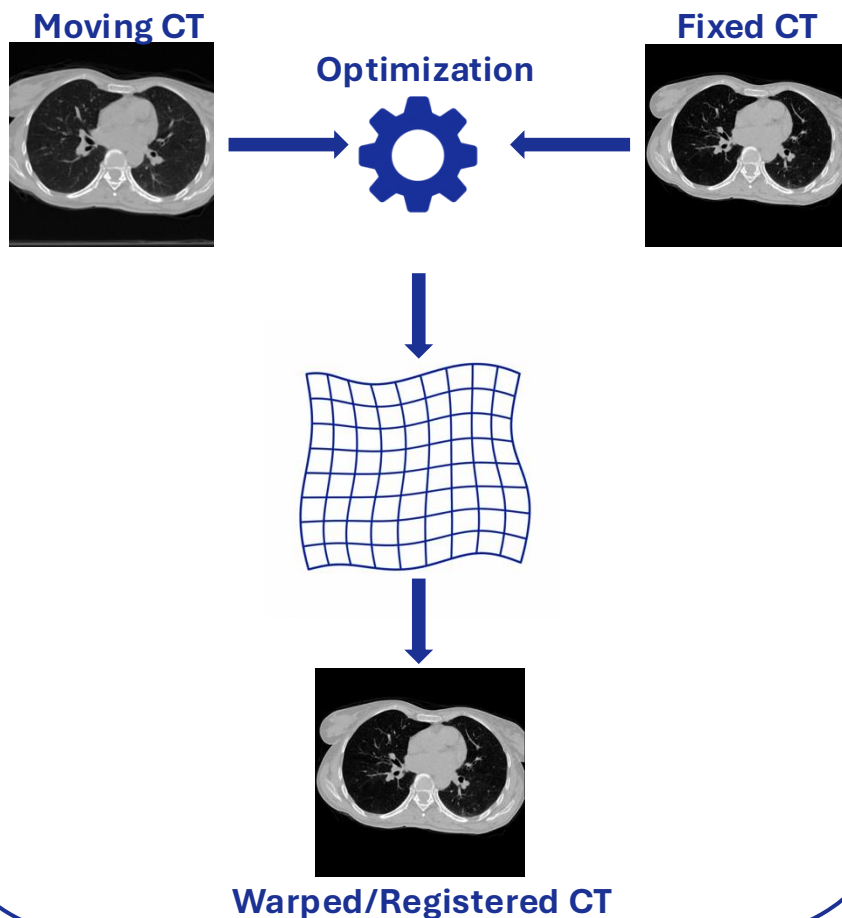
WHAT WE NEED



From registration to deformation generation



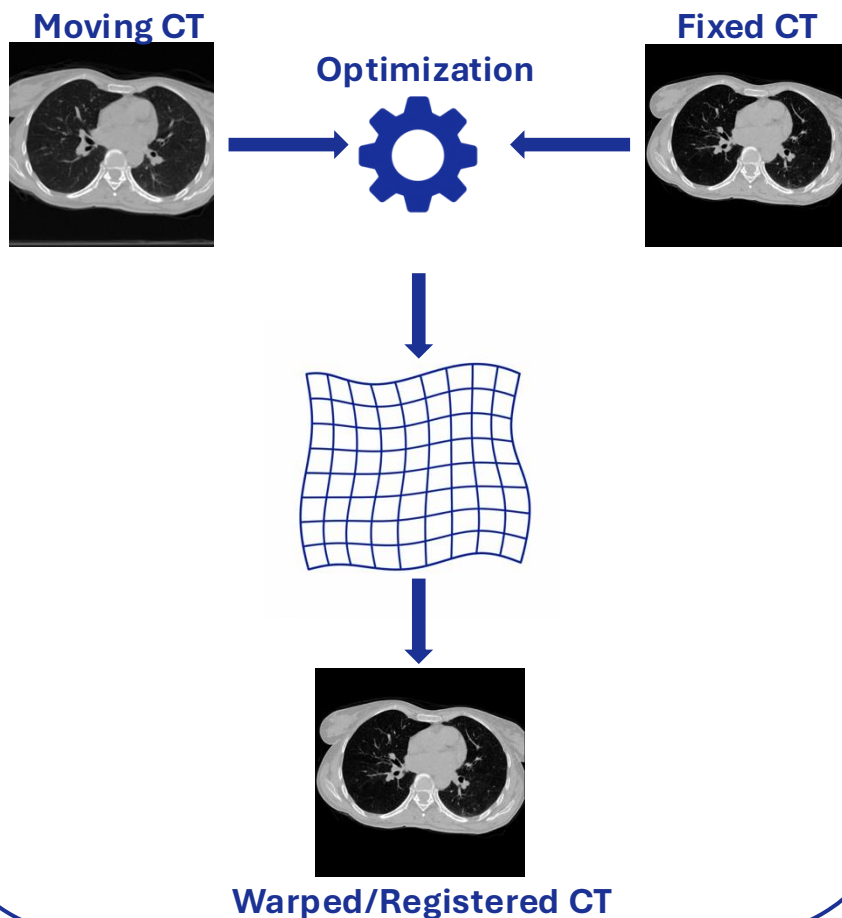
CLASSICAL REGISTRATION



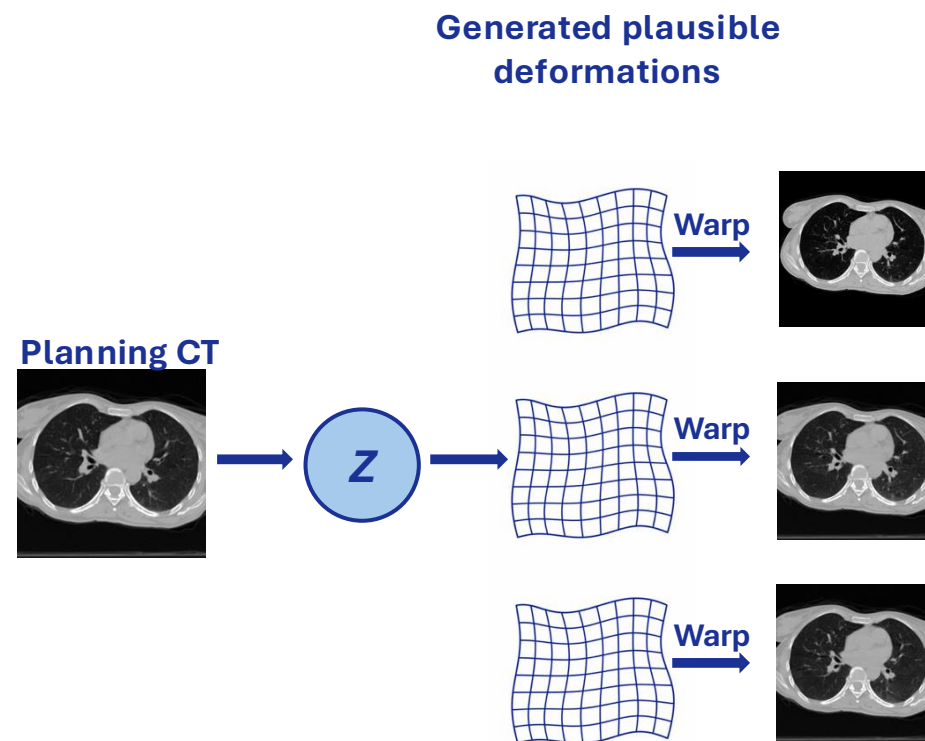
From registration to deformation generation



CLASSICAL REGISTRATION



DEFORMATION GENERATION



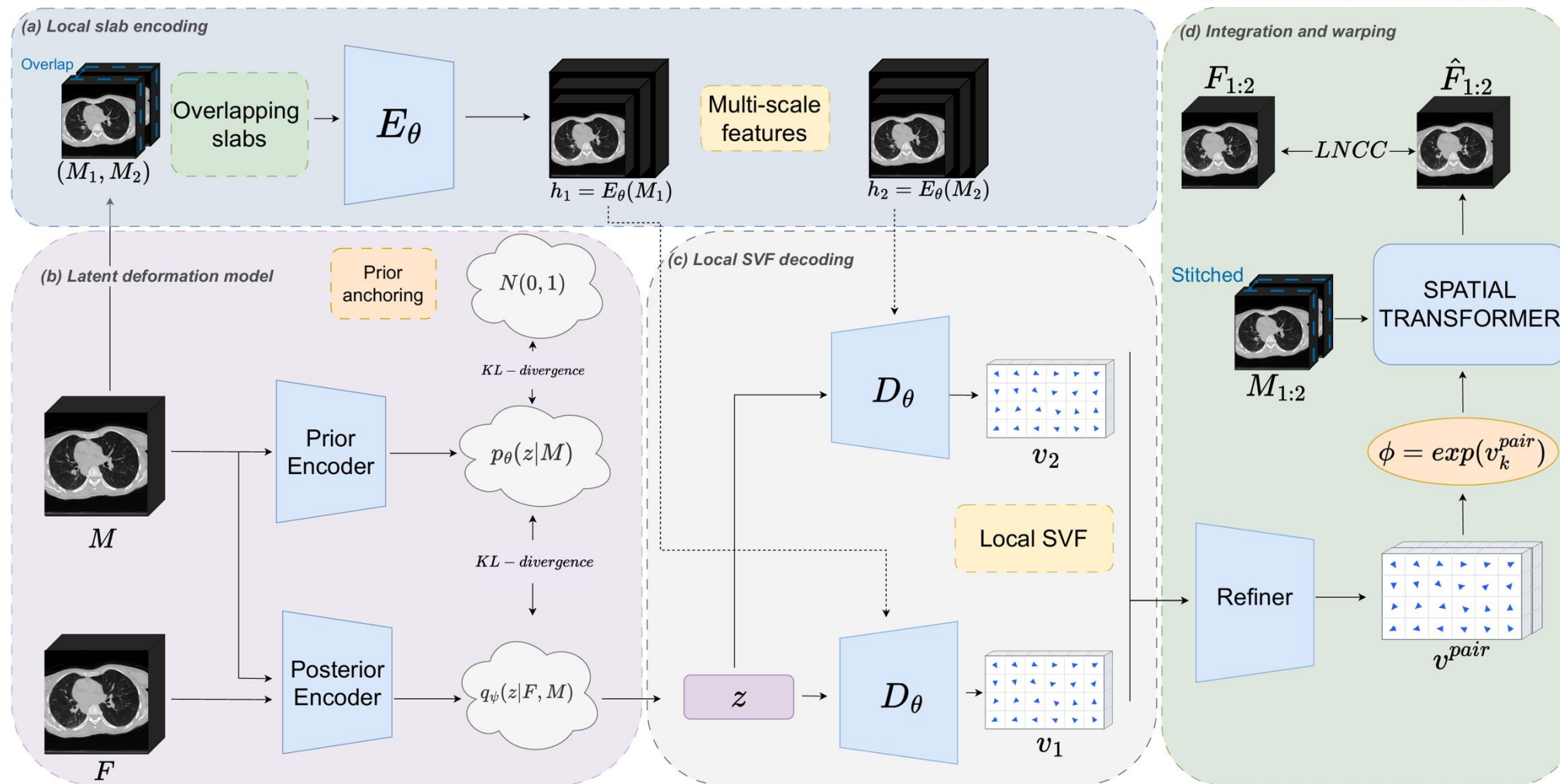
Related work:

Pastor-Serrano et al., PMB, 2023.

Smolders et al., PMB, 2024.

METHODOLOGY

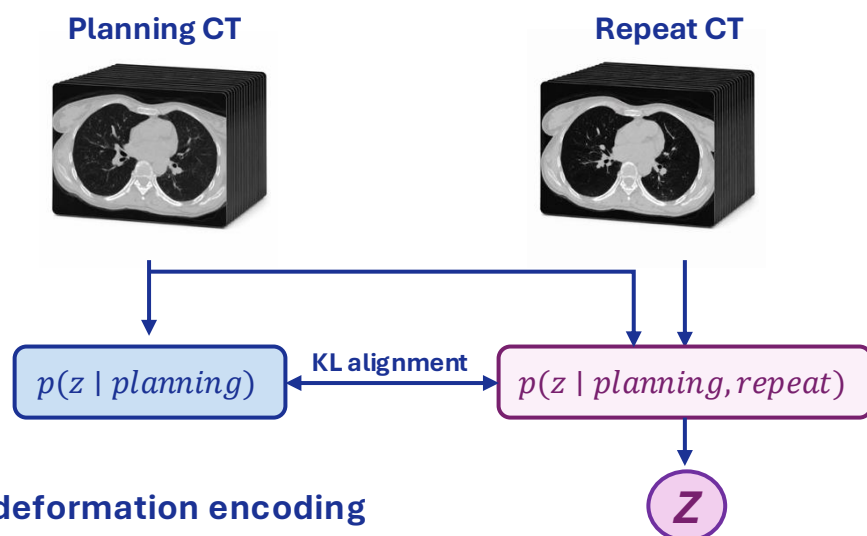
SynthRCT overview



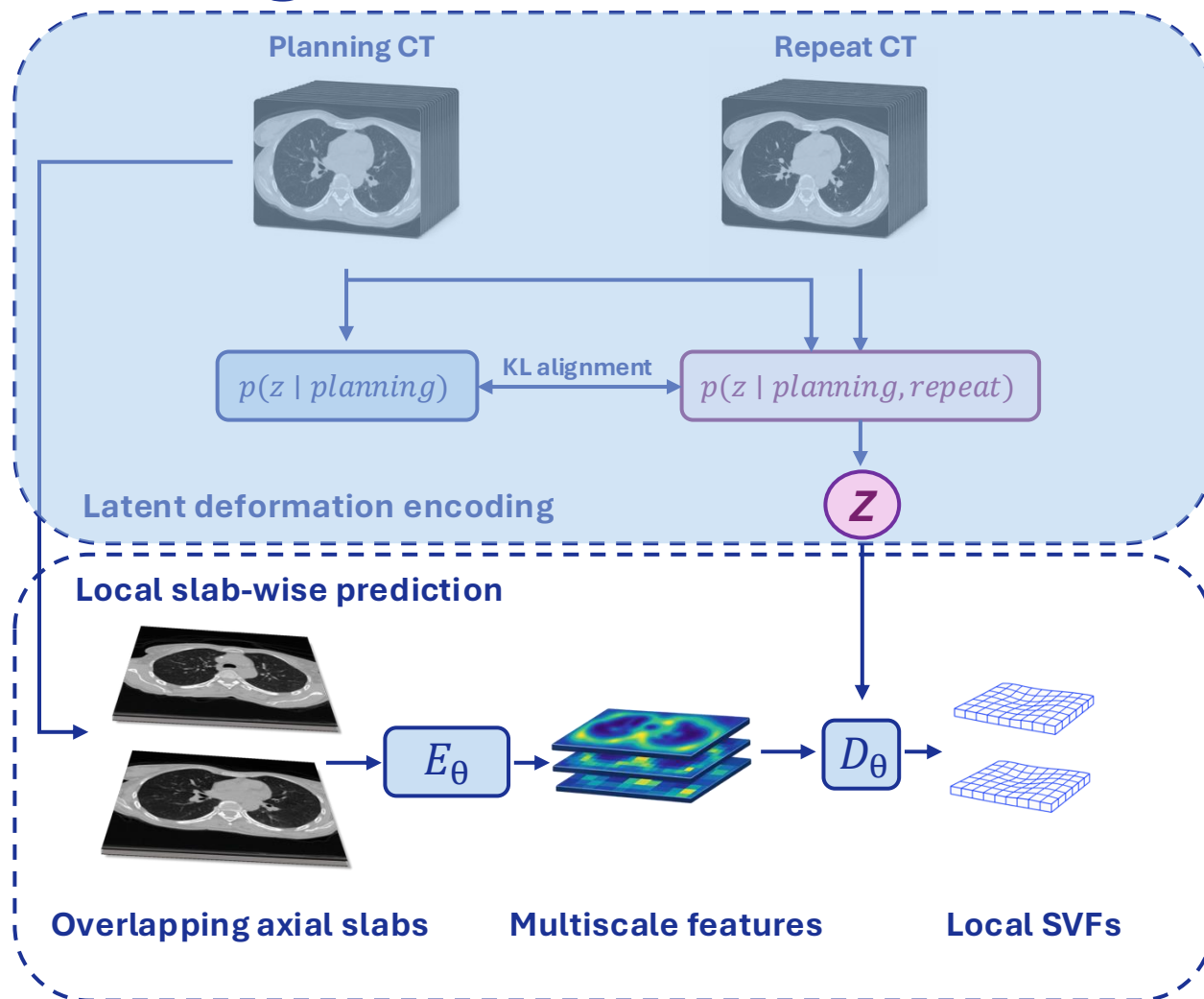
MICCAI2026



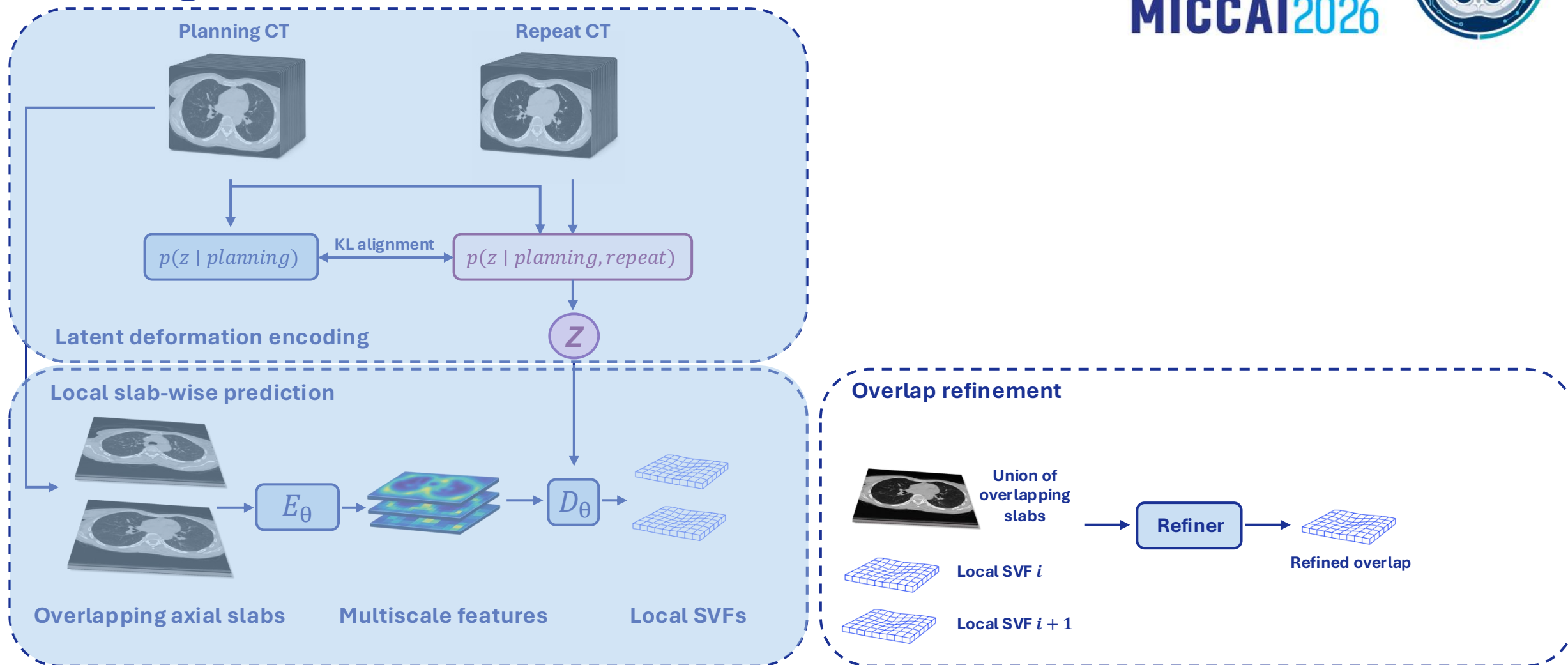
Training



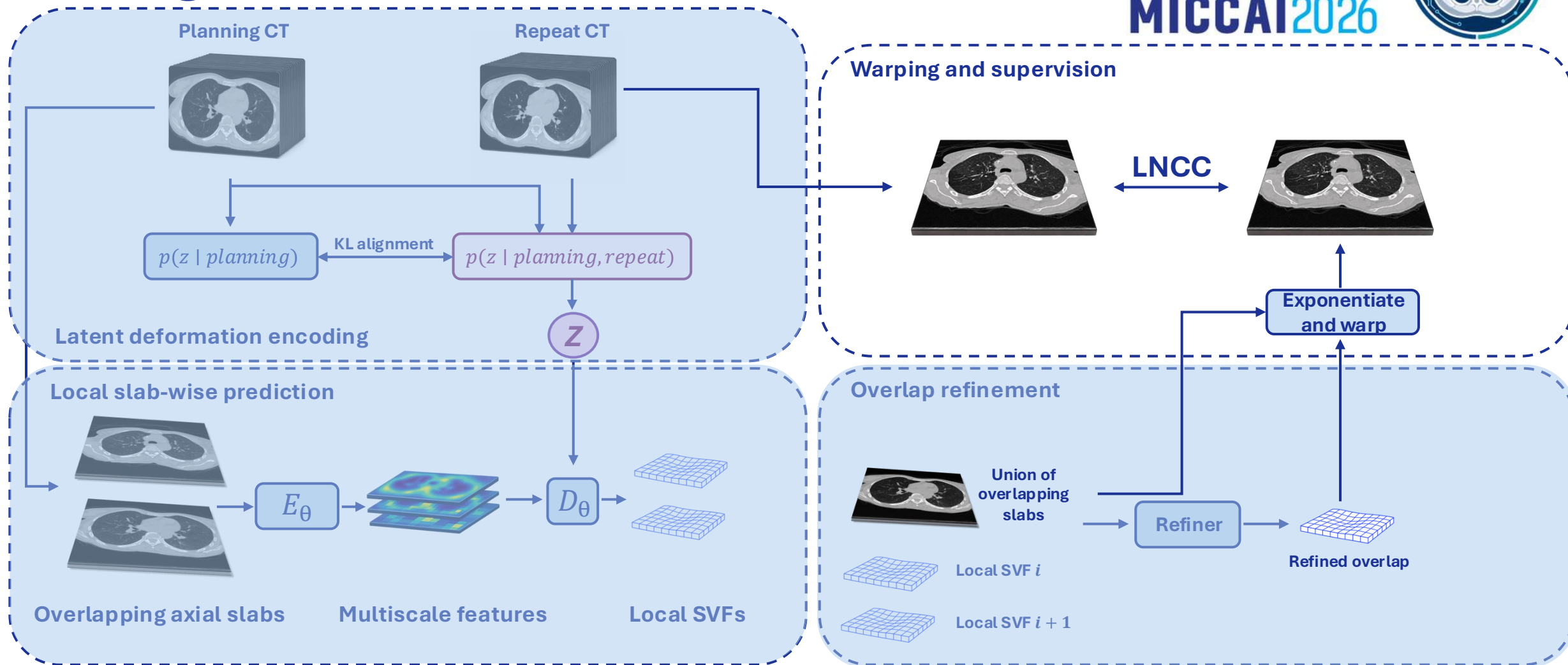
Training



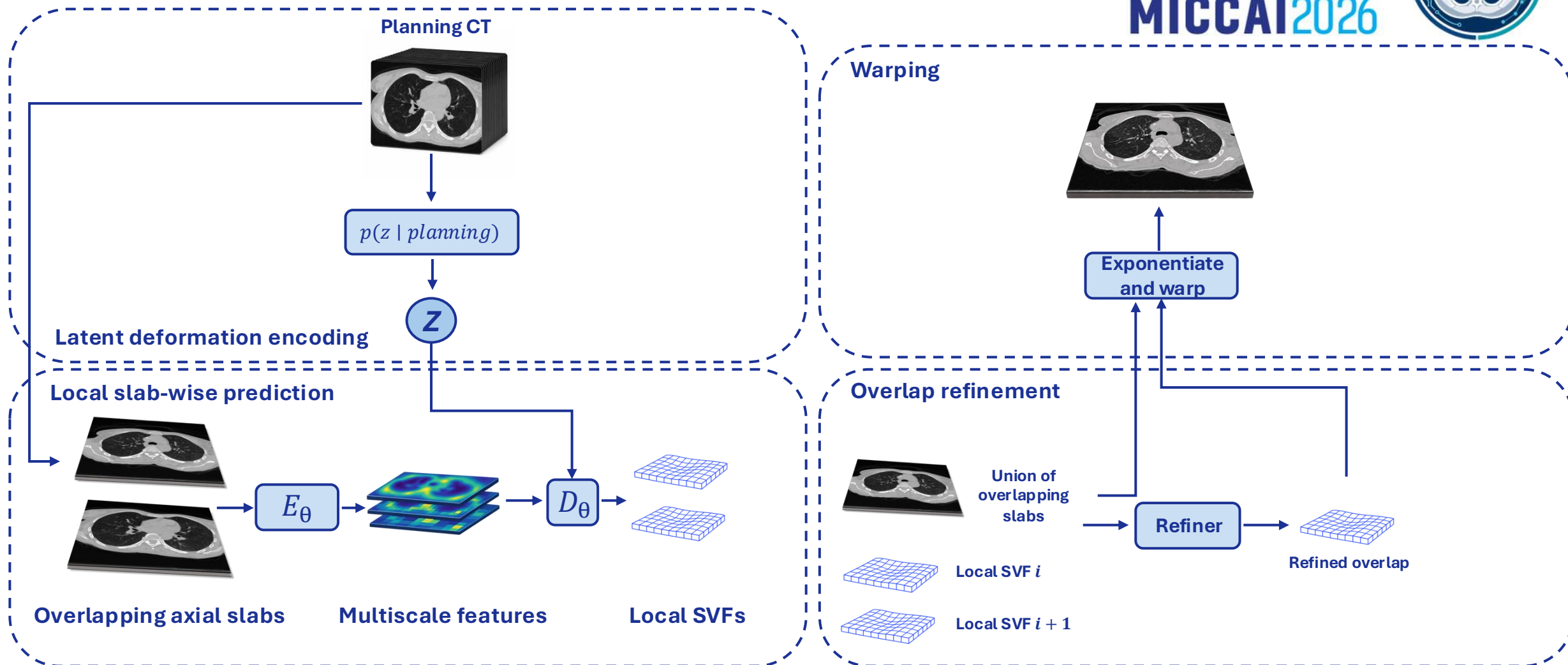
Training



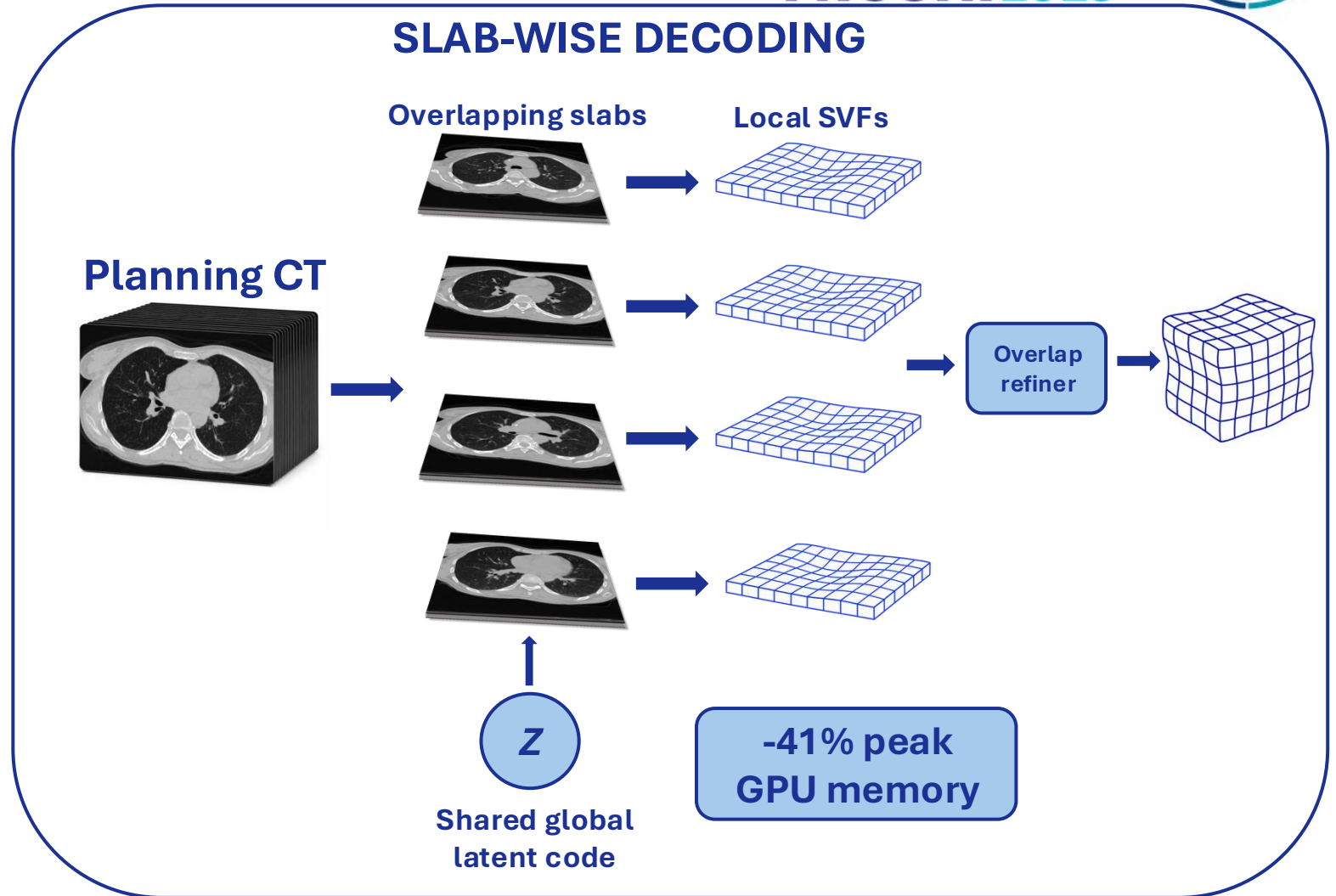
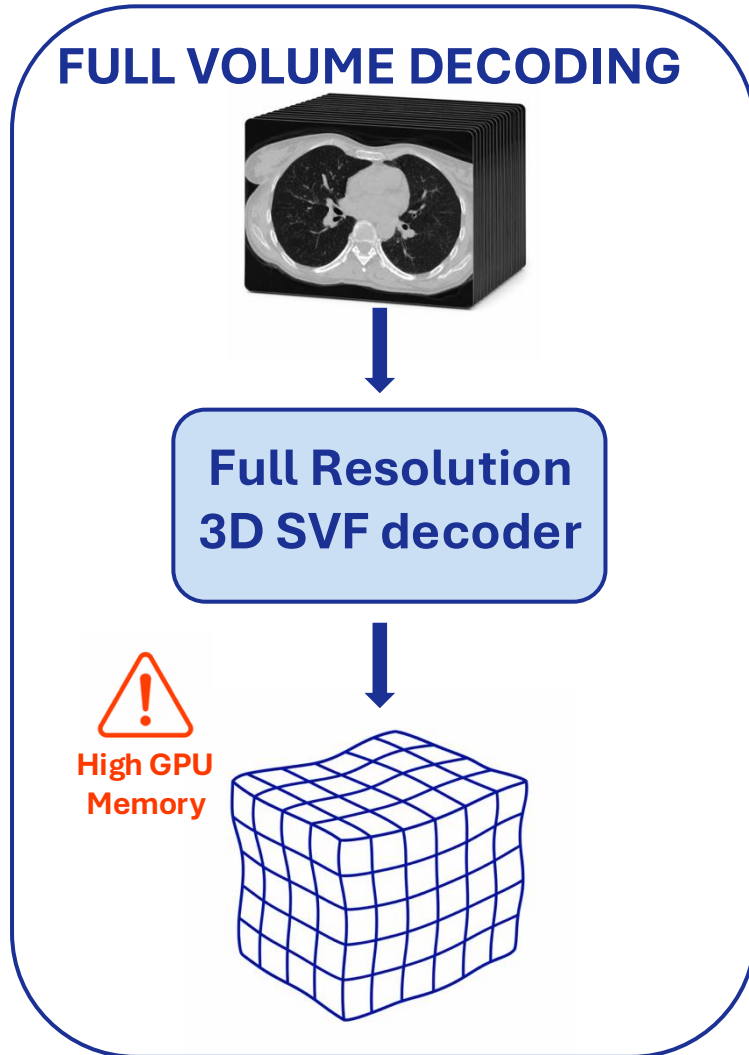
Training



Inference



Scaling deformation generation to full 3D CT



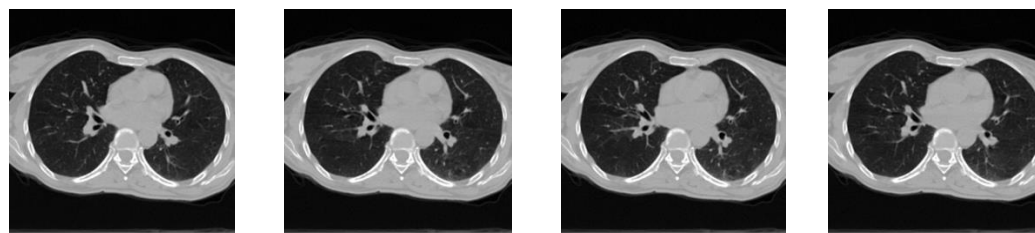
Dataset and experimental scope



DIR4DCT

-  10 patients
-  10 respiratory phases
-  1.5 mm isotropic res.
-  256 x 256 x 207 voxels
-  75 landmarks

Example phases



————— Respiratory cycle —————→



We can model and test:

- Patient-specific respiratory cycle.
- Topology-preserving diffeomorphic motion.
- Agreement with observed motion distribution.

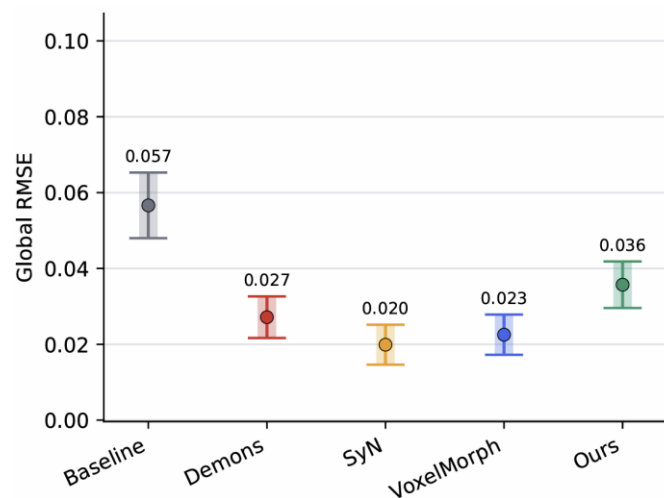


Not yet covered:

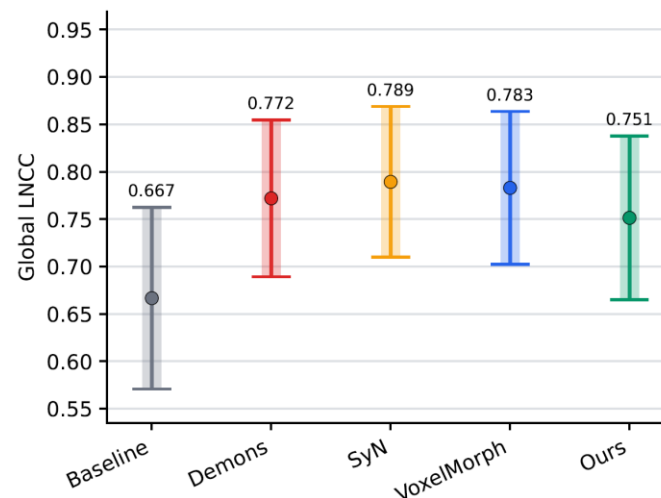
- Tumour shrinkage.
- Weight loss.
- Broader anatomical variations.

RESULTS

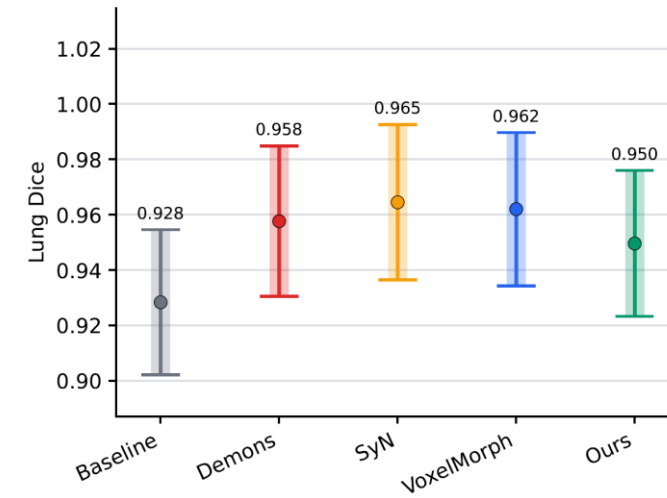
General deformation quality



RMSE



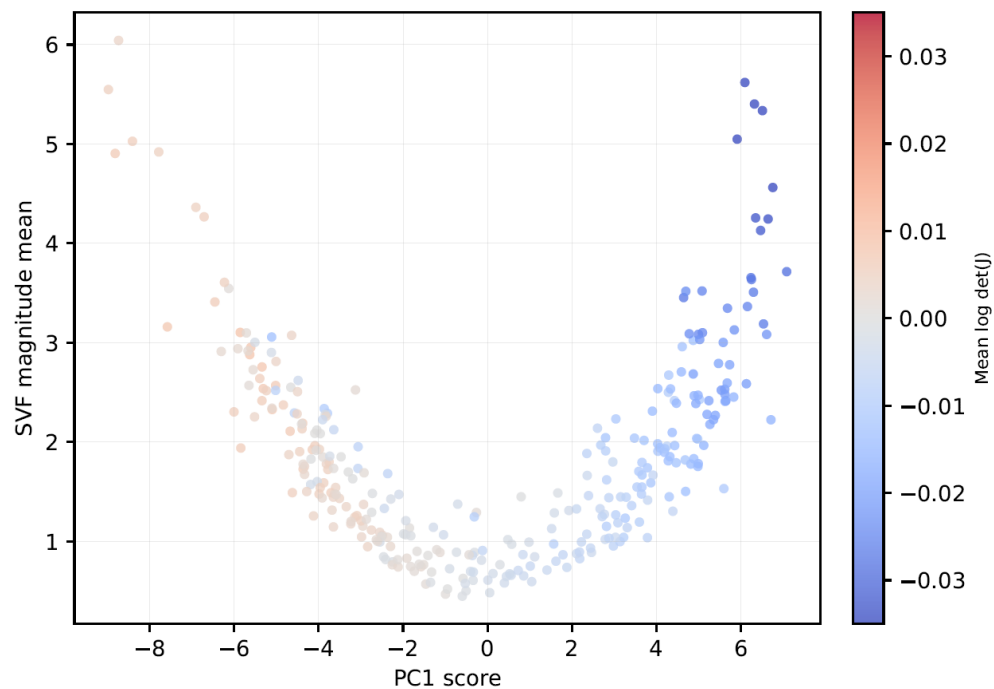
LNCC



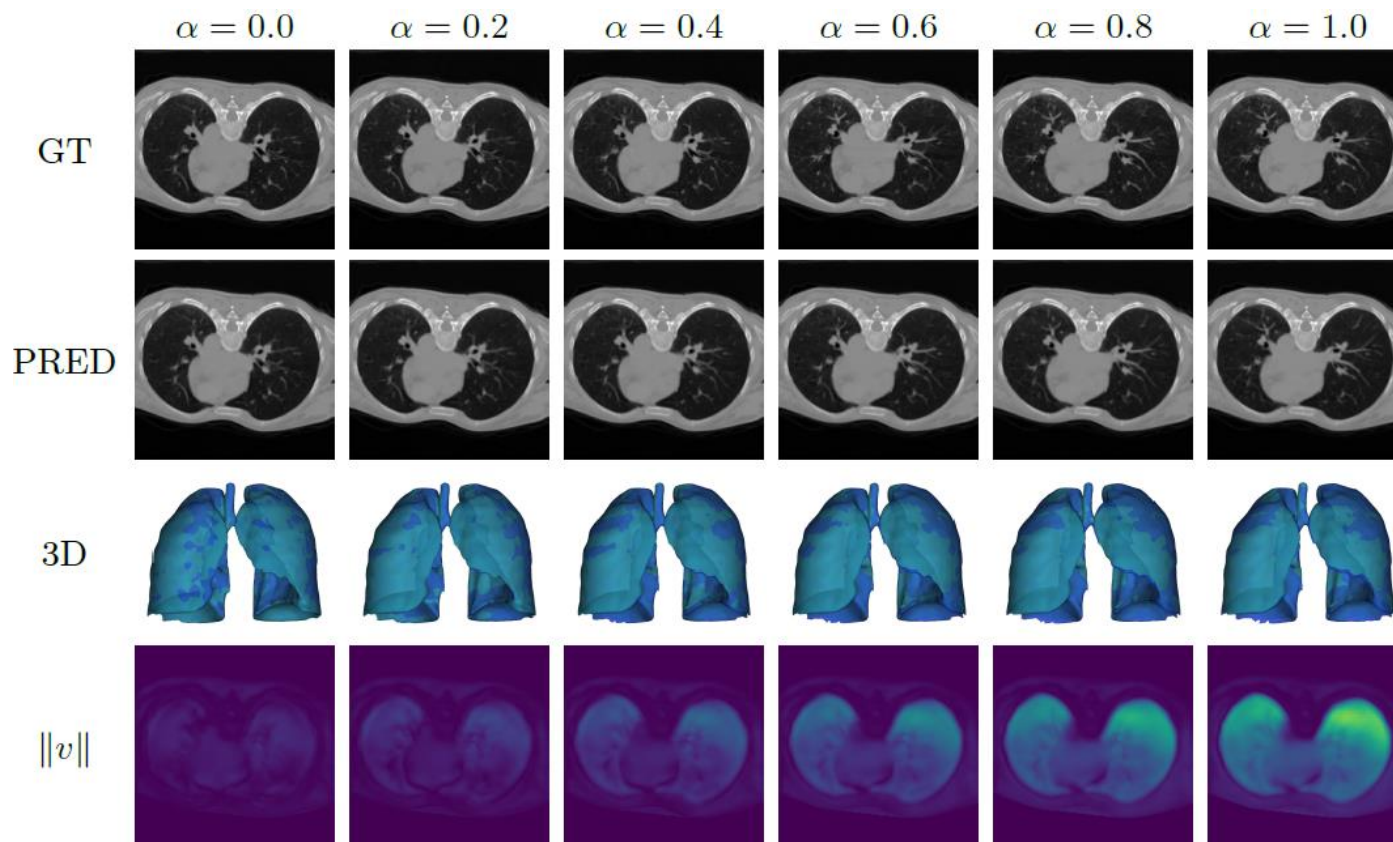
DICE

Sanity check: generated deformations improve alignment — outperforming registration methods is not the goal.

Latent space analysis



Latent direction captures deformation magnitude

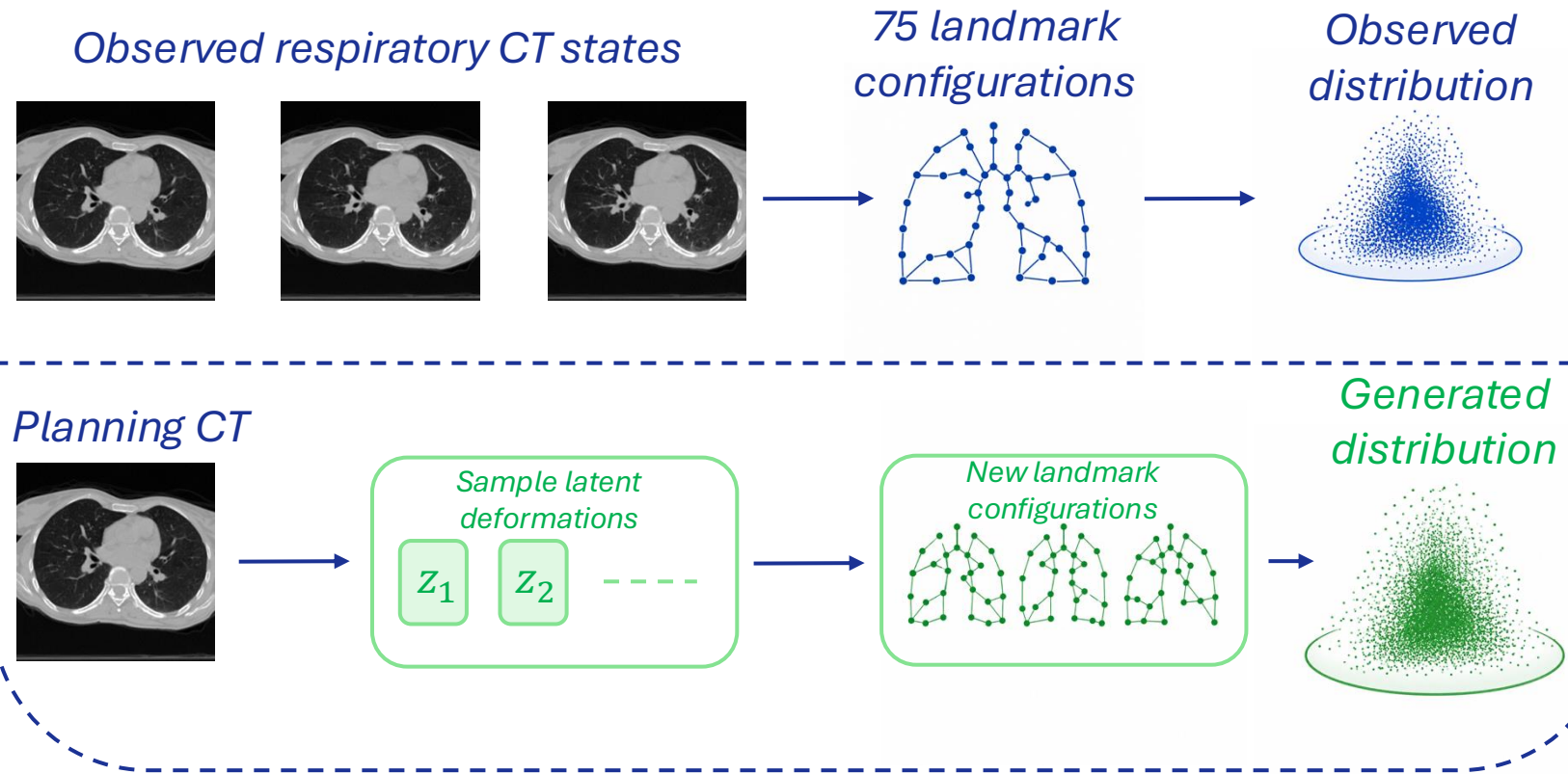


Smooth interpolation between respiratory states

Landmark distribution agreement



Evaluation setup

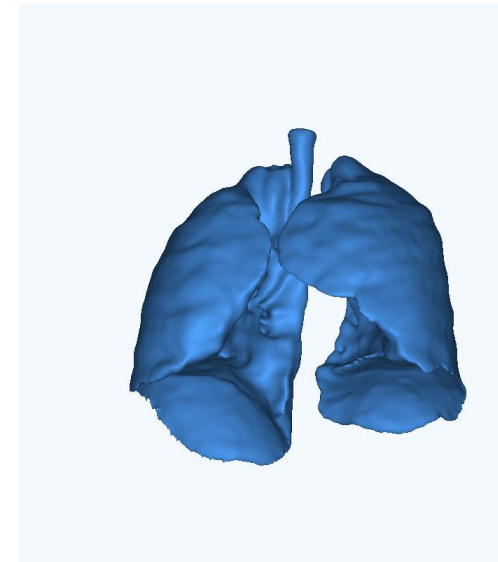
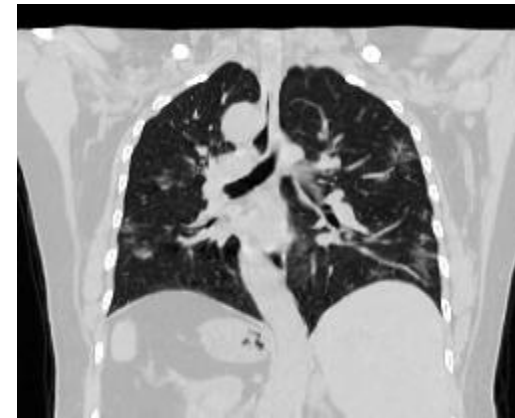
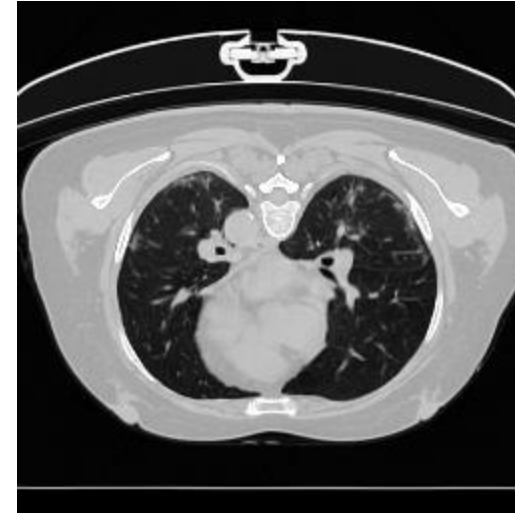


Landmark distribution evaluation (held-out patient)			
	Metric	Value (mm)	Interpretation
	Energy distance D_E	1.44	Global distribution discrepancy
	Wasserstein-1 W_1	2.61	Global distribution discrepancy
	Coverage d_{cov}	1.69	Real states close to generated
	Precision d_{prec}	1.78	Generated states close to real
	Spread ratio ρ_{spread}	0.98	Matched motion variability

CONCLUSIONS

Conclusions

- SynthRCT learns patient-specific distributions of plausible anatomical deformations from a single planning CT.
- Slab-wise SVF decoding enables full-Resolution 3D generation with substantially lower GPU memory requirements.
- The learned latent space is structured and smooth, supporting meaningful interpolation between respiratory states.
- Generated deformations reproduce observed respiratory variability at millimeter scale, while remaining smooth and topology preserving.



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Tomas Guija-Valiente

THANK YOU FOR YOUR ATTENTION!